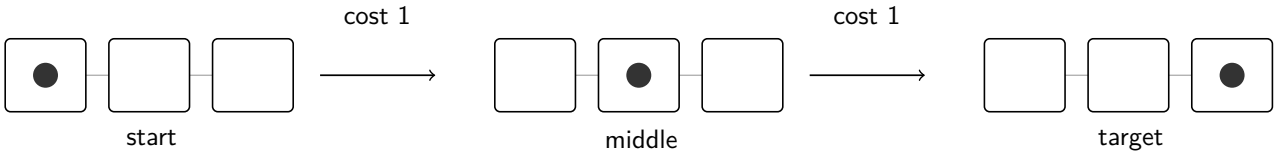
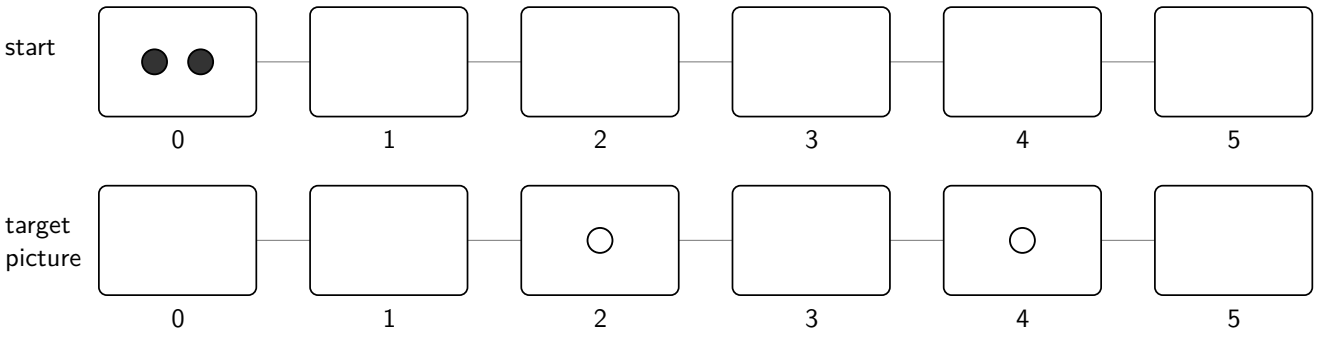
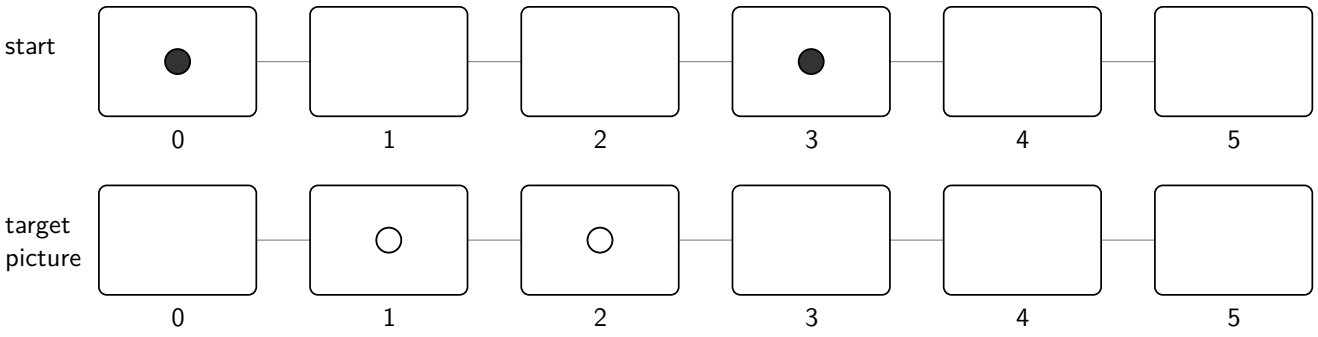
Move one counter at a time to a neighboring bin. Each crossing costs 1. Bins hold any number of counters, and counters may pass through occupied bins. Any counter may fill any target place. Only start counters move; the target picture stays fixed. Keep one record of your moves at a time so you can count the work afterward.

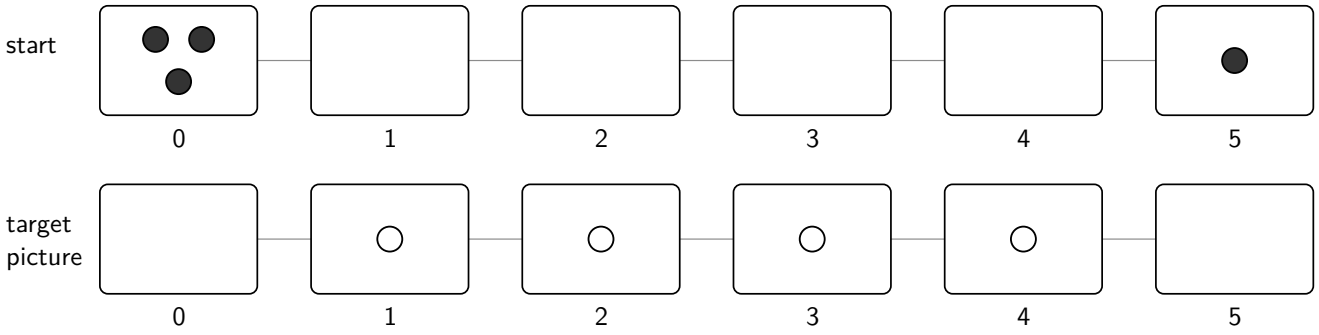



**Problem 1:** Match each target using the least work you can. Find two best move records for each pair. Changing the order of moves counts as different.

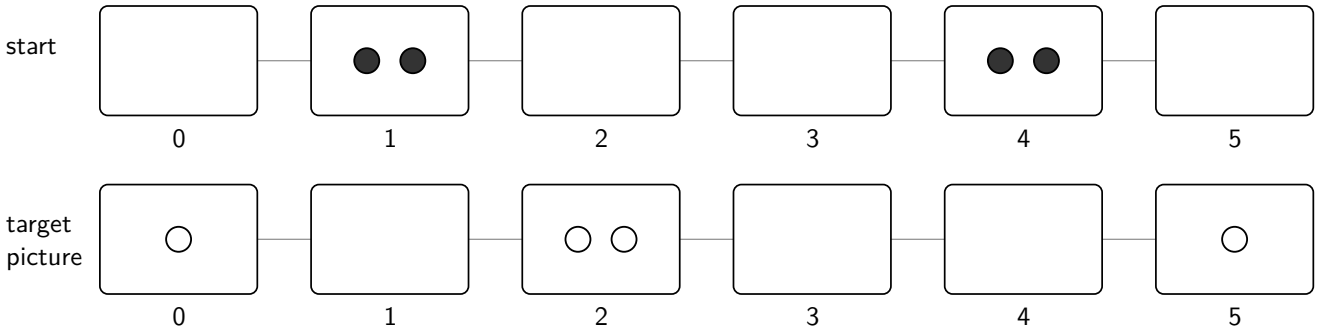


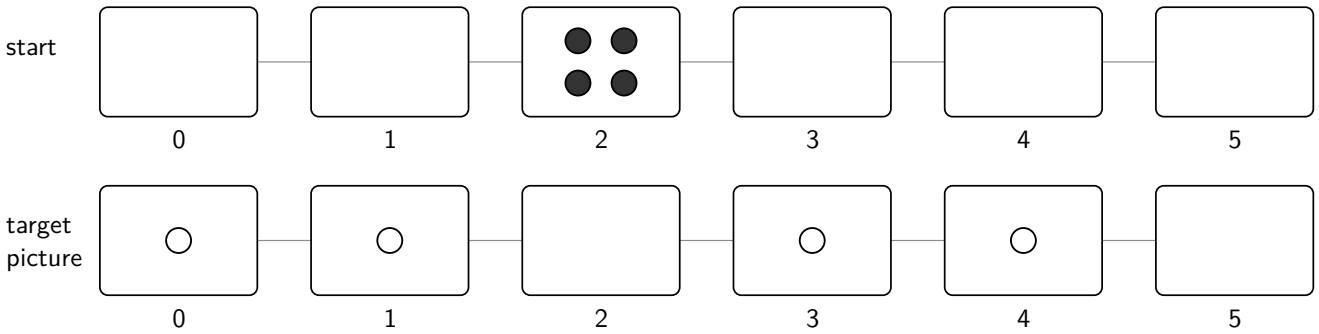
**Problem 2:** Make each start match its target. How little work can you use?




work: _____

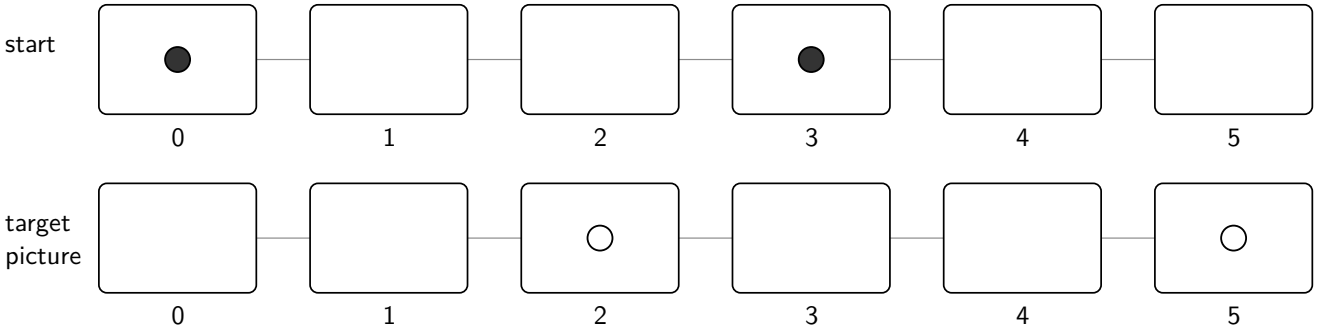


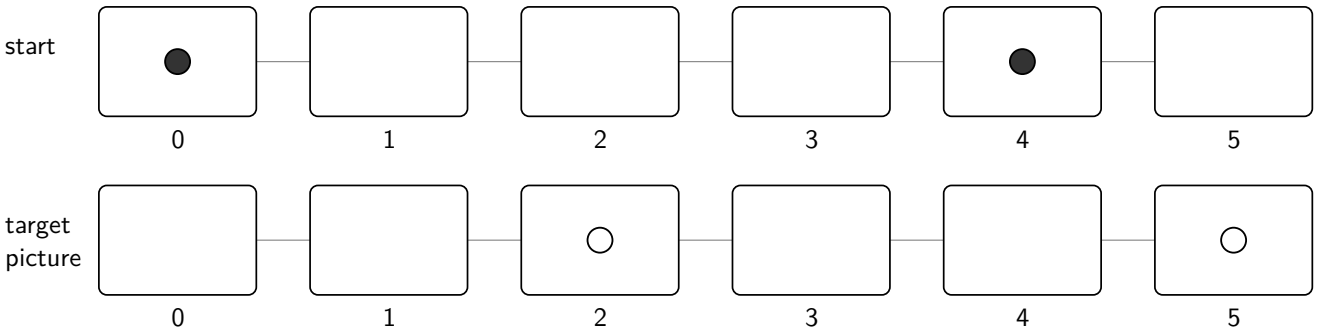

work: _____




work: _____

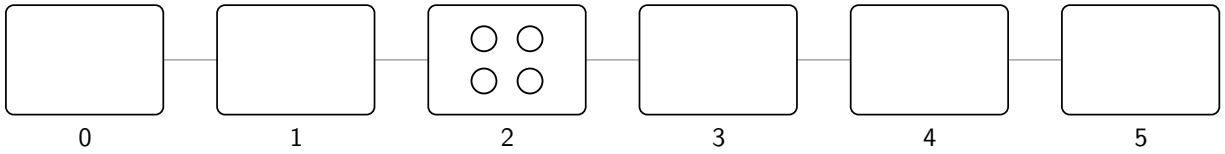
**Problem 3:** A delivery takes one counter that has not yet been delivered all the way to one target place that has not yet been filled. Once delivered, that counter stays there. Try this rule: always make the shortest remaining delivery. If deliveries tie, choose either. For each pair, can another plan use less work?



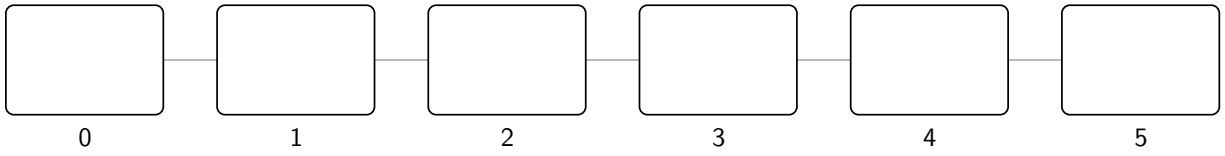



**Problem 4:** Use four counters and keep this target fixed. Find different starts whose least work is 4. Then find starts whose least work is 7.

target picture

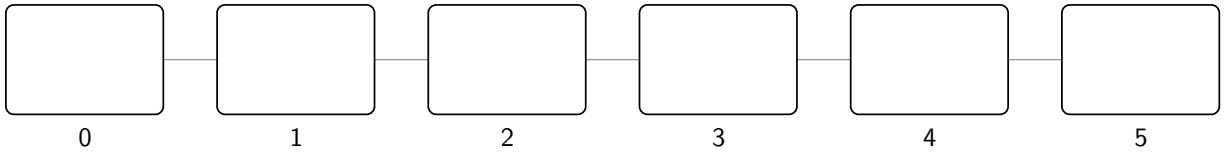



start



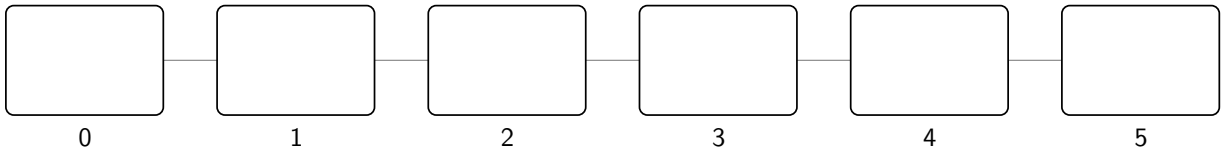

least work: _____

start



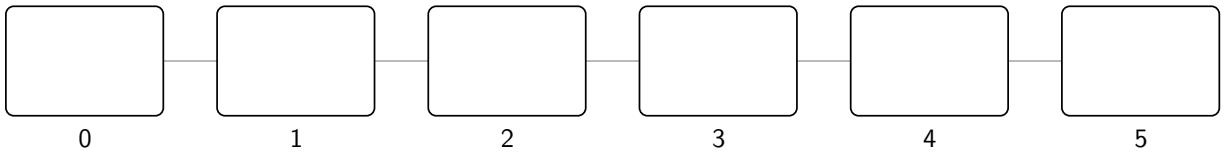

least work: _____

start



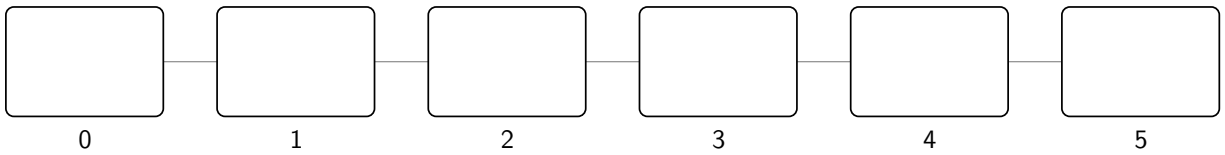

least work: _____

start




least work: _____

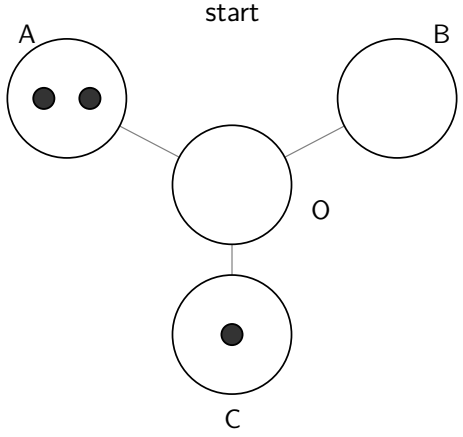
start



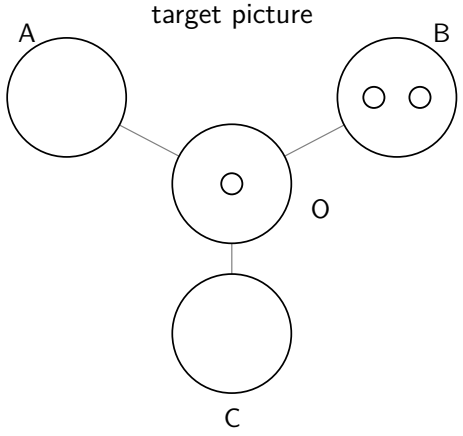

least work: _____

**Problem 5:** Move counters along the roads to match each target. One counter crossing one road costs 1. Find the least work for each pair. Then draw a new start and target with three counters each whose least work is 6.

start

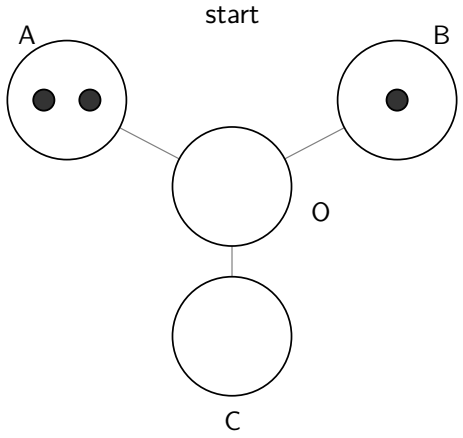



target picture

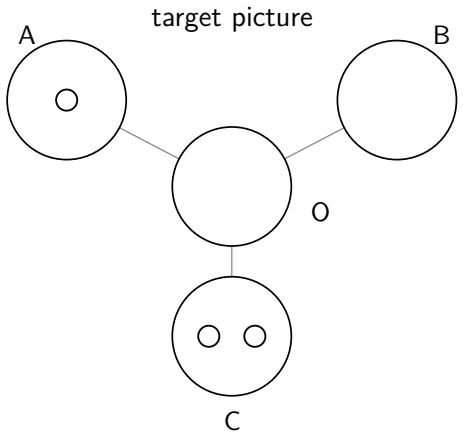



work: _____

start

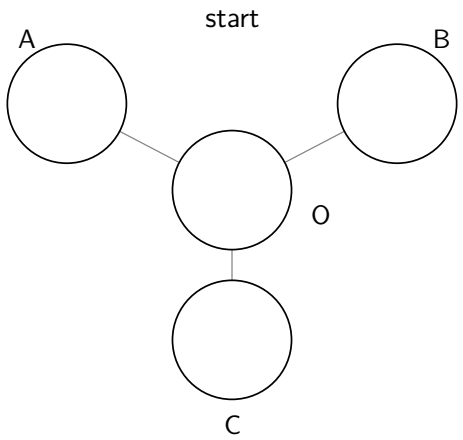



target picture

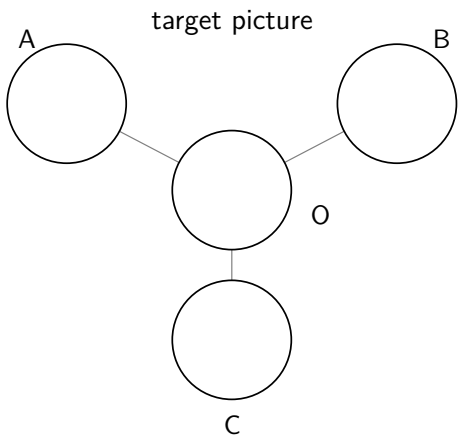



work: _____

start

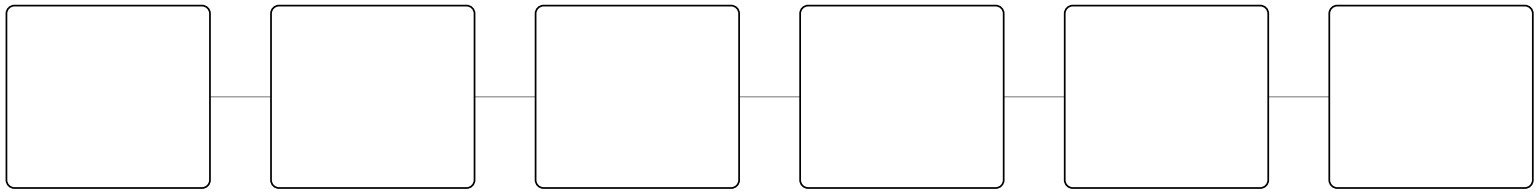



target picture

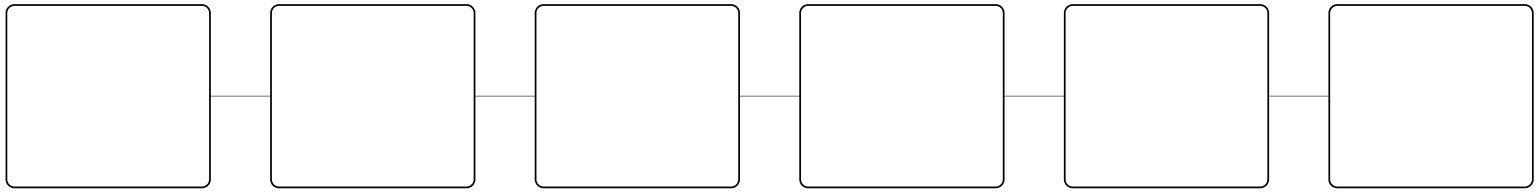


**Problem 6:** One player draws a target with four counters. The other puts four counters on the start row so the least work is exactly 5. Check it, then swap. Can a target make this impossible?

start




target picture




work: _____